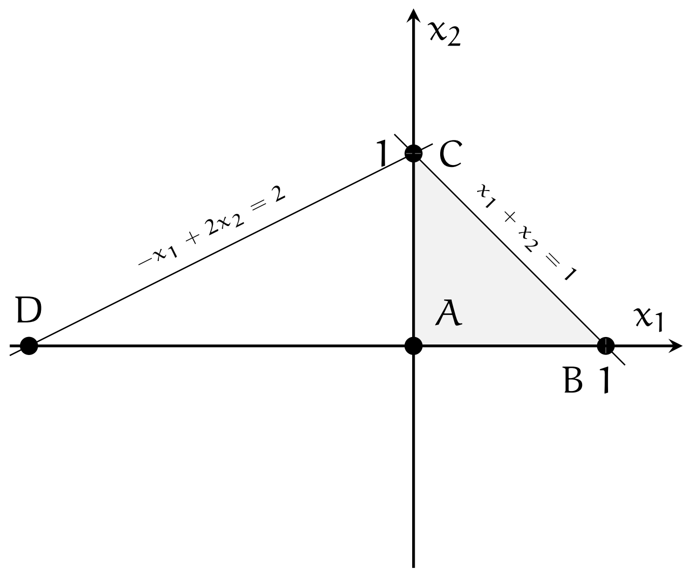
Polyhedron P The polyhedron P is represented below:



The corresponding polyhedron in standard form is obtained by introducing non negative slack variables, and rewriting the constraints in the form of equality constraints. We obtain a polyhedron in $\mathbb{R}^4$:

$$\left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4 | x_1 + x_2 + x_3 = 1, -x_1 + 2x_2 + x_4 = 2, x_1, x_2, x_3, x_4 \geq 0 \right\}.$$

The equality constraints in matrix form are

$$Ax = b, \text{ where } A \in \mathbb{R}^{2\times 4},\ b \in \mathbb{R}^2,\ x \in \mathbb{R}^4,$$
$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ -1 & 2 & 0 & 1 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

We have a problem with $n = 4$ variables and $m = 2$ equality constraints. There are 6 ways to select 2 basic variables out of 4.

1. Basic solution where x_1 and x_2 are in the basis:

$$B = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}; B^{-1} = \begin{pmatrix} \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 0 \\ 1 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

The basic solution is feasible and corresponds to the point $C = (0, 1)$. It is a degenerate solution as one of the basic variables (x_1) is zero. There are 5 active constraints, two equality constraints, and three inequality constraints: $x_1 \geq 0$, $x_3 \geq 0$, $x_4 \geq 0$.

2. Basic solution where x_1 and x_3 are in the basis

$$B = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}; B^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} -2 \\ 3 \end{pmatrix}; x = \begin{pmatrix} -2 \\ 0 \\ 3 \\ 0 \end{pmatrix}.$$

The basic solution is not feasible, as $x_1 < 0$. It corresponds to the point $D = (-2, 0)$.

3. Basic solution where x_1 and x_4 are in the basis

$$B = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}; B^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 3 \end{pmatrix}; x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 3 \end{pmatrix}.$$

   The basic solution is feasible and corresponds to the point $B = (1, 0)$. It is not degenerate as no basic variables is zero.

4. Basic solution where x_2 and x_3 are in the basis

$$B = \begin{pmatrix} 1 & 1 \\ 2 & 0 \end{pmatrix}; B^{-1} = \begin{pmatrix} 0 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

   The basic solution is feasible and corresponds to the point $C = (0, 1)$. It is a degenerate solution as one of the basic variables (x_3) is zero. There are 5 active constraints, two equality constraints, and three inequality constraints: $x_1 \geq 0$, $x_3 \geq 0$, $x_4 \geq 0$.

5. Basic solution where x_2 and x_4 are in the basis

$$B = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}; B^{-1} = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

   The basic solution is feasible and corresponds to the point $C = (0, 1)$. It is a degenerate solution as one of the basic variables (x_4) is zero. There are 5 active constraints, two equality constraints, and three inequality constraints: $x_1 \geq 0$, $x_3 \geq 0$, $x_4 \geq 0$.

6. Basic solution where x_3 and x_4 are in the basis

$$B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; B^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 2 \end{pmatrix}.$$

   The basic solution is feasible and corresponds to the point $A = (0, 0)$. It is not degenerate as no basic variable is zero.

   In presence of degeneracy, a vertex may correspond to multiple feasible basic solutions. Here the vertex

$$\begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}$$

   corresponds to the bases (x_1, x_2), (x_2, x_3) and (x_2, x_4). Five constraints are active and, in dimension 4, four constraints are sufficient to identify a vertex.

   In the original polyhedron, it corresponds to the point $C = (1, 0)$. At that point, three constraints are active:

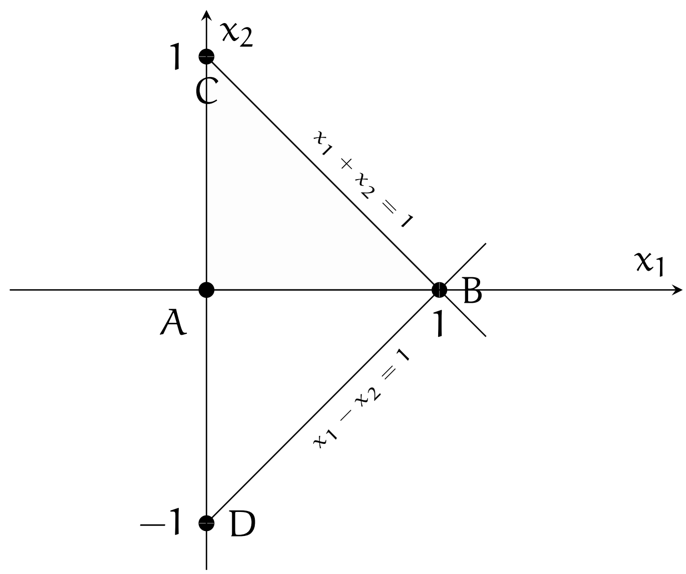
- $x_1 + x_2 \leq 1$,
- $-x_1 + 2x_2 \leq 2$,
- $x_2 \geq 0$.

In dimension 2, two constraints are sufficient to identify a vertex.

Polyhedron Q We represent the polyhedron

$$Q = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \in \mathbb{R}^2 | x_1 + x_2 \leq 1, x_1 - x_2 \leq 1, x_1 \geq 0, x_2 \geq 0 \right\}$$

in the following figure:



The corresponding polyhedron in standard form is

$$\left\{ \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in \mathbb{R}^4 | Ax = b, x \geq 0 \right\},$$

with

$$A = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 1 & -1 & 0 & 1 \end{pmatrix}, b = \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

with the variables x_3 and x_4 being slack variables. Each basic solution is obtained by selecting 2 variables out of 4 to be in the basis. There is total of 6 possible selections of basic variables.
The matrix $B = (A_{j1}, ..., A_{jm})$ is composed of columns $j_1, ..., j_m$ of the matrix A.

1. Basic solution with x_1 and x_2 in the basis ($j_1 = 1, j_2 = 2$).

$$B = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}; B^{-1} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

This basic solution is feasible and corresponds to point $B = (1, 0)$ in the figure.

2. Basic solution with x_1 and x_3 in the basis $(j_1 = 1, j_2 = 3)$.

$$B = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}; B^{-1} = \begin{pmatrix} 0 & 1 \\ 1 & -1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

This basic solution is feasible and also corresponds to point $B = (1, 0)$.

3. Basic solution with x_1 and x_4 in the basis $(j_1 = 1, j_2 = 4)$

$$B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}; B^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 0 \end{pmatrix}; x = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}.$$

This basic solution is feasible and corresponds to point $B = (1, 0)$.

4. Basic solution with x_2 and x_3 in the basis $(j_1 = 2, j_2 = 3)$.

$$B = \begin{pmatrix} 1 & 1 \\ -1 & 0 \end{pmatrix}; B^{-1} = \begin{pmatrix} 0 & -1 \\ 1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} -1 \\ 2 \end{pmatrix}; x = \begin{pmatrix} 0 \\ -1 \\ 2 \\ 0 \end{pmatrix}.$$

This basic solution is not feasible because $B^{-1}b \ngeq 0$. It corresponds to point $D = (0, -1)$ in the figure.

5. Basic solution with x_2 and x_4 in the basis $(j_1 = 2, j_2 = 4)$

$$B = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}; B^{-1} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 1 \\ 0 \\ 2 \end{pmatrix}.$$

This basic solution is feasible and corresponds to point $C = (0, 1)$ in the figure.

6. Basic solution with x_3 and x_4 in the basis $(j_1 = 3, j_2 = 4)$

$$B = B^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}; x_B = B^{-1}b = \begin{pmatrix} 1 \\ 1 \end{pmatrix}; x = \begin{pmatrix} 0 \\ 0 \\ 1 \\ 1 \end{pmatrix}.$$

This basic solution is feasible and corresponds to point $A = (0, 0)$ in the figure.

In summary, out of the six basic solutions, three are degenerate, and they correspond to the same vertex, identified by five active constraints: two equality constraints, and three inequality constraints: $x_2 \geq 0$, $x_3 \geq 0$, and $x_4 \geq 0$. In the original polyhedron Q, among the six basic solutions,

- one corresponds to the vertex A, where the two constraints $x_1 \geq 0$ and $x_2 \geq 0$, associated with the non basic variables, are active,
- three correspond to the vertex B, where the **three** constraints $x_2 \geq 0$, $x_1 + x_2 \geq 1$, and $x_1 - x_2 \geq 1$ are active,
- one corresponds to the vertex C, where the two constraints $x_1 \geq 0$ and $x_1 + x_2 \leq 0$, associated with the non basic variables, are active,
- one is infeasible, and corresponds to point D, where the two constraints $x_1 \geq 0$ and $x_1 - x_2 \leq 1$ are active.

The three basic solutions associated with vertex B are degenerate. From an algebraic point of view, one of the basic variables is equal to zero. From a geometrical point of view, more than two constraints are active.